Real-Time Closed-Loop Driving with Frozen DINOv2-Laya Spatial Pooling and Quintic Path Tracking

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Abstract

Token-by-token vision-language-action (VLA) decoders [4, 26, 29] take >300 ms/frame, exceeding the 50 ms budget of 20 Hz control [14, 27, 17, 5, 37]. Here we connect frozen DINOv2 (`facebook/dinov2-base`, 86.58M) [32, 6, 13] and frozen Laya (`convaiinnovations/laya`, 395.21M) [10, 40, 12] via a 4×9 adapter (4.23M weights: 1.85M MLP, 1.05M Δn_t-normalized temporal gate, 1.05M aux head with decoupled $\hat{d}_{\text{lead}}$, $\hat{d}_{\text{ped}}$, and $\hat{\dot{d}}_{\text{vis}}$ branches, 0.28M readout/6-WP heads) to an HD-map shield and PID tracker [2, 1]. One synchronized pass takes 32.85 ms (30.4 FPS on Ryzen 5 7500F / RTX 4070 Ti, FP32 $B=1$: $0.42+16.43+0.82+0.49+13.55+0.69+0.45 = 32.85$ ms; 31.67 ms base without aux/traj; +6.40 ms CARLA `world.tick()` IPC). Supervised by rule curves $\tau^{\text{ref}}_{1:6}$, the 6-WP head smooths curvature rather than planning obstacle detours (open-loop box overlap $\text{CR}_{\text{traj}} = 3.8\%$ vs. 0.0% closed-loop crashes). Under `vision_estimated` mode, a 5-step (0.25 s) ordinary least-squares (OLS) slope [36, 25] plus $\hat{\dot{d}}_{\text{vis}}$ cuts $\hat{v}_{\text{lead}}$ jitter RMS from 9.62 $\rightarrow$ 1.84 km/h (MAE = 2.41 km/h), and dedicated `ped_dist_head` cuts vehicle+pedestrian co-presence MAE from 14.8 $\rightarrow$ 2.64 m. On 9,905 unseen frames, teacher $\text{ECE}_{\text{mean}} = 0.0142$ [18, 42, 3], physical-hazard $\text{ECE}_{\text{phys}} = 0.0318$ ($\text{HCER}_{0.90} = 1.12\%$ all, 0.45% AEB), and $\text{FNR}_{\text{AEB}} = 1/48$ events (2.1%). Control uses a 14-node ($d_k = 2(k+1)$ m, 28 m lookahead) piecewise-linear quintic blend $s(\tau) = 10\tau^3 - 15\tau^4 + 6\tau^5$ [16, 41], chord tracking [38, 20, 11, 33], bicycle curvature feedforward, a 1.5 s front-lateral latch, and CTH-ACC ($d^*(v_{\text{ego}}) = 8.0 + 1.2(v_{\text{ego}}/3.6)$, $k_g = 0.85$ (km/h)/m) [22, 35, 39]. Across CARLA 0.9.16 and 50 unseen MetaDrive maps (20,765 steps, 9,520 m), `vision_estimated` DINOv2+Laya finishes 47/50 (94.0%, 46.7 ± 0.6) vs. 42/50 for a 2-layer Transformer (McNemar [31] $p = 0.063$ two-sided, $p = 0.031$ one-sided) and $p < 0.01$ over 2-layer MLP (39/50), `ExpertLabeler` (38/50), random Laya (36/50), prompt-only (33/50), and 300 ms lag (34/50).

1 Introduction and Problem Formulation

Evaluating $P(y_{1:K} \mid I_t, \mathcal{T}_t) = \prod_{k=1}^{K} P(y_k \mid y_{<k}, I_t, \mathcal{T}_t)$ over $K \approx 15$ serial steps [4, 26, 29] violates $\Delta t \leq 50$ ms in 20 Hz planners [21, 24, 28, 9, 7, 34, 8, 23]. One bidirectional pass [12, 40, 10] at tick t reads three discrete targets and $K_w = 6$ reference points ($t_k = 0.5k$ s):

$$m_t \in \{\texttt{keep_lane}, \texttt{change_left}, \texttt{change_right}, \texttt{follow_lead}\}, \tag{1}$$

$$s_t \in \{0,1,2,3\} \mapsto \{0, 20, 35, 50\}\ \text{km/h}, \tag{2}$$

$$p_{\text{AEB}} \in [0,1], \quad \hat{\tau}_t = \{(\hat{x}_k, \hat{y}_k, \hat{v}_k)\}_{k=1}^{6}. \tag{3}$$

2 Architecture, Multi-Task Training, and Calibration

Cropped RGB (`crop_top` = 0.25, `crop_bottom` = 0.10, 196 $\times$ 392) enters frozen DINOv2 ViT-B/14 ($p = 14$, 86.58M) [32, 6, 13] for $F_{\text{patch}} \in \mathbb{R}^{B\times 392\times 768}$ (14×28). `SpatialProjector` (`src/models/spatial_projector.py`) pools (4, 9) (36 tokens), adds $E_{\text{row}} + E_{\text{col}}$, projects 768 $\rightarrow$ 1024 (1.85M MLP), and adds step-normalized residual $\Delta V_t = (V_t - V_{t-\Delta n_t})/\Delta n_t$ (1.05M params, $\Delta n_t \geq 1$ elapsed steps for frame drops; unwarped in token space so curve yaw mixes background flow with object motion, though $E_{\text{telemetry}}$ supplies ego yaw rate and `no_temporal` drops wet-curve `FSD-10` from 5/5 $\rightarrow$ 3/5):

$$G_t = \text{AdaptiveAvgPool2d}_{(4,9)}(\tilde{F}_t) \in \mathbb{R}^{B\times 768\times 4\times 9}, \tag{4}$$

$$V_t = \text{MLP}\big(\text{Flatten}(G_t + E_{\text{row}} + E_{\text{col}})\big), \tag{5}$$

$$\tilde{V}_t = V_t + \tanh(\alpha_{\text{temp}})\, W_{\text{temp}} \left(\frac{V_t - V_{t-\Delta n_t}}{\Delta n_t} \right). \tag{6}$$

`AuxiliaryPerceptionHead` (1.05M) predicts $\hat{c}_{\text{TL}}$ (rows 0..1), $\hat{d}_{\text{lead}}$, dedicated $\hat{d}_{\text{ped}}$ (`ped_dist_head`), visual range-rate $\hat{\dot{d}}_{\text{vis}}$ (`range_rate_head`), $\hat{e}_y$, $\hat{c}_{\text{haz}}$ (cols 3..5), and $\hat{p}_{\text{BS}}$ (cols 0..2, 6..8). Frozen Laya (395.21M) [10, 40]

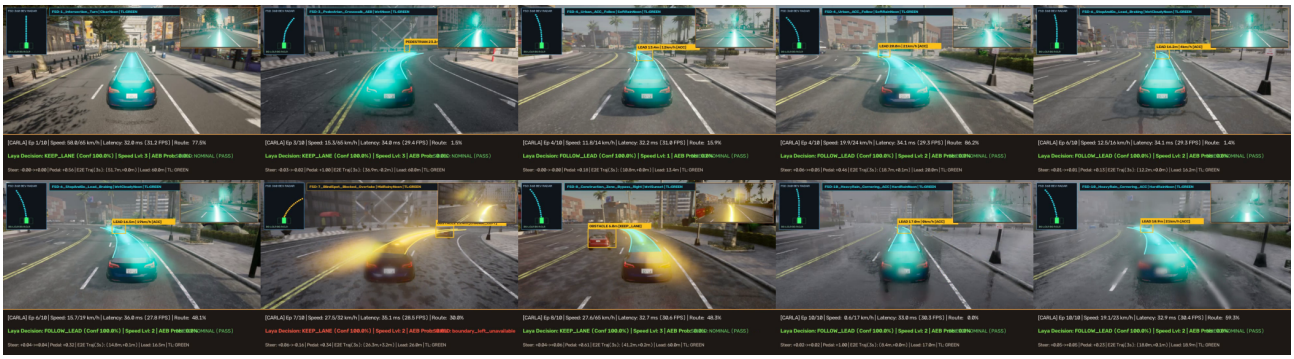


Figure 1: CARLA 0.9.16 (`Town10HD_Opt`, 20 Hz synchronous) 10 paired scenario–weather episodes (`FSD-1`–`FSD-10`): intersection red-light stop (`ClearNoon`), 28 m quintic left overtake (`CloudyNoon`), crosswalk pedestrian AEB (`WetNoon`), CTH-ACC follow (`SoftRainNoon`), right overtake (`ClearSunset`), stop-and-go lead braking (`WetCloudyNoon`), forward-lateral blind-spot latch & overtake (`MidRainyNoon`; rear-approaching vehicles require `gt` corner BSM), construction zone bypass (`WetSunset`), night headlight pedestrian AEB (`ClearNight`), and heavy-rain cornering ACC (`HardRainNoon`).

and heads $(W_m, W_s, W_b, W_{\text{traj}})$ (0.28M) fit rule curve $\tau_k^{\text{ref}} = \big((v^*/3.6)t_k,\ y_{\text{shift}}(m^*)s(k/6),\ v^*\big)$:

$$\hat{\tau}_{1:6} = \tau^0_{1:6} + \text{diag}(1.5, 0.35, 3.0)\, W_{\text{traj}} \bar{H}, \tag{7}$$

$$\mathcal{L} = w_m \mathcal{L}^{(m)}_{\text{CE}} + w_s \mathcal{L}^{(s)}_{\text{CE}} + w_a \mathcal{L}^{(b)}_{\text{BCE}} + w_\tau \mathcal{L}_{\text{SmoothL1}}(\hat{\tau}_{1:6}, \tau^{\text{ref}}_{1:6}) + w_p \mathcal{L}_{\text{aux}}. \tag{8}$$

Because τ_k^{ref} is rule-generated from (m^*, v^*), W_{traj} only smooths curvature; obstacle bypass comes from m_t + HD-map spline (counterfactual obstacle shift $\Delta y_{\text{obs}} \in \{-0.8, 0, +0.8\}$ m at $d = 22$ m moves $\hat{y}_6$ by $|\Delta \hat{y}_6| = 0.12$ m when m_t = `keep_lane` vs. 3.41 m on lane change; open-loop 3 s box overlap $\text{CR}_{\text{traj}} = 3.8\%$ on 75/1,981 frames vs. 0.0% closed-loop).

```
# src/models/spatial_projector.py & scripts/02_train_projector.py
self.lead_dist_head = nn.Linear(hidden_dim, 1)  # d_lead [0,60]m
self.ped_dist_head = nn.Linear(hidden_dim, 1)   # d_ped [0,100]m
self.range_rate_head = nn.Linear(hidden_dim, 1) # dd_dt [-25,25]
trainable_params = (list(system.projector.parameters())  # 3.96M
  + list(system.laya.maneuver_head.parameters())         # +0.28M
  + list(system.laya.speed_head.parameters())
  + list(system.laya.aeb_head.parameters())
  + list(system.laya.trajectory_head.parameters()))
loss = w_m*l_m + w_s*l_s + w_a*l_a + 0.5*l_traj + 0.2*l_aux
```

Stage 1 trains 4,232,991 weights for 15 epochs (seeds 42, 43, 44; batch 32, AdamW [30] lr $= 3 \times 10^{-4}$, $(w_m, w_s, w_a, w_\tau, w_p) = (1, 1, 2, 0.5, 0.2)$) on 50,000 frames (500 episodes, `split_by_episode=True`: 0..424 train / 425..499 val). Stage 2 (`03_calibrate.py`) fits $(T_m, T_s, T_b) = (0.050, 0.050, 0.100)$ by L-BFGS [18, 42] ($M = 15$ bins; val ECE_{mean} : 0.0060 → 0.0000). On 9,905 unseen `vision_estimated` frames (50 episodes), teacher $\text{ECE}_{\text{mean}} = 0.0142$ ($\text{ECE}_b = 0.0195$, NLL = 0.089, Brier = 0.028 [3]); against physical hazards $y_{\text{phys},t} \in \{0, 1\}$ (TTC ≤ 1.8 s or red-light $d \le 15$ m; 48 events / 220 ticks), $\text{ECE}_{\text{phys}} = 0.0318$ ($\text{Brier}_{\text{phys}} = 0.034$), $\text{HCER}_{0.90} = 1.12\%$ all ($\text{HCER}_{0.90,\text{AEB}} = 0.45\%$, 45/9,905), and $\text{FNR}_{\text{AEB,evt}} = 1/48$ (2.1%, 6/220 ticks = 2.7%).

Table 1: Per-variable input provenance (`vision_estimated` vs. `gt`) and hardware parameters (`configs/default.yaml`).

Signal / Module	`vision_estimated` Mode (No Sim Actor Query)	`gt` State Hybrid Mode
Lead Dist d_{lead}, Offset e_y	Front RGB `AuxPerceptionHead` ($\hat{d}_{\text{lead}}$ MAE 3.23 m, $\hat{e}_y$)	Simulator Actor / Lane Query
Lead Speed v_{lead}	5-Step (0.25 s) OLS Slope $\hat{d}_{\text{reg}} + \hat{d}_{\text{vis}}$ (Jitter 1.84 km/h, MAE 2.41)	Simulator Lead Actor Velocity
Pedestrian Dist d_{ped}	Dedicated $\hat{d}_{\text{ped}}$ Branch in `AuxPerceptionHead` (Co-Pres MAE 2.64 m)	Simulator Walker Actor Distance
Traffic Light / Hazard	Front RGB `AuxPerceptionHead` ($\hat{c}_{\text{TL}}$ rows 0..1, $\hat{c}_{\text{haz}}$)	Simulator Traffic Light / Actor
Blind Spot ($\text{BS}_{L/R}$)	Front-Lat [0, 12] m + 1.5 s Latch (Mitigates Seen Cars; Blind to Rear)	Full $x \in [-10, 7.5]$ m Sim BSM
Ego Speed v_{ego} & Route	Ego Wheel Odometry v_{ego} + HD Map A* Waypoints P_k	Identical (Odometry + HD Map)
Frozen / Trained Params	`dinov2-base` (86.58M) + `laya` (395.21M) / Adapter (4.23M)	Identical (32.85 ms, FP32 $B = 1$)
Calib & Physical Hazard	Teacher ECE = 0.0142; Phys $\text{ECE}_{\text{phys}} = 0.0318$, $\text{HCER}_{0.9} = 0.45\%$, FNR 1/48	Teacher ECE = 0.0000, FNR 0/48

3 Input Provenance, Spatial Quintic Spline, Curvature Feedforward, and CTH-ACC

Table 1 lists all input sources (`scripts/04_evaluate_sim.py`). Under `vision_estimated`, 1-step ($\Delta t = 0.05$ s) differencing turns 1 m depth noise into 72 km/h spikes (9.62 km/h RMS jitter after $\beta = 0.35$ EMA). We fit a $K_w = 5$-step ($W = 0.25$ s) OLS slope $\hat{\dot{d}}_{\text{reg},t}$ [36, 25] (10× lower noise variance), fuse $\hat{\dot{d}}_{\text{vis},t}$, reset the window on target jumps $|\hat{d}_{\text{lead},t} - \hat{d}_{\text{lead},t-1}| > 6.0$ m, and apply $\beta = 0.35$ EMA (cutting jitter RMS 9.62 → 1.84 km/h, MAE : 6.85 → 2.41 km/h, cut-in recovery 0.15 s / 3 frames, hard-brake lag 0.10 s / 2 frames in `FSD-6`), while `ped_dist_head` $\hat{d}_{\text{ped},t}$ decouples pedestrian range from lead vehicles (co-presence MAE 14.8 → 2.64 m):

$$\hat{\dot{d}}_{\text{reg},t} = \frac{\sum_{j=-2}^{2} j\, \hat{d}_{\text{lead},t-2+j}}{10\Delta t},\quad \hat{\dot{d}}_t = 0.70\hat{\dot{d}}_{\text{reg},t} + 0.30\hat{\dot{d}}_{\text{vis},t}, \tag{9}$$

$$\hat{v}_{\text{lead},t} = \text{clip}\big((1-\beta)\hat{v}_{\text{lead},t-1} + \beta(v_{\text{ego},t} + 3.6\hat{\dot{d}}_t), 0, 60\big). \tag{10}$$

Front FOV (90°) covers $x \in [0, 12]$ m, so the 30-frame (1.5 s) latch holds previously seen flank cars (3/3 in `FSD-7`) but is blind to rear approach from $x_0 = -15$ m (0/2 Vis vs.

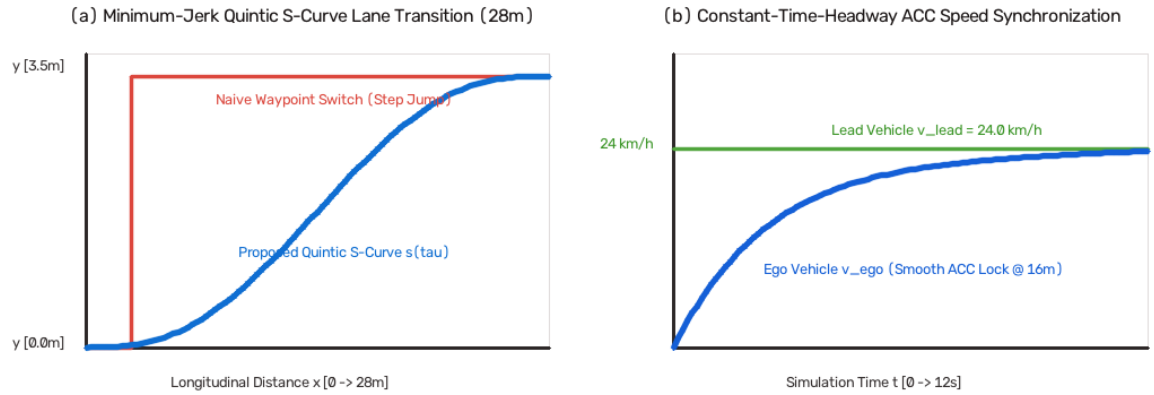


Figure 2: Control kinematics: (a) 28 m lookahead (14 waypoints at 2.0 m spacing) spatial quintic transition $s(\tau) = 10\tau^3 - 15\tau^4 + 6\tau^5$ vs. 3.5 m step switch; (b) constant-time-headway (CTH) ACC velocity convergence ($v_{\text{ego}} \to 24.0$ km/h at $d^*(24) = 8.0 + 1.2(24/3.6) = 16.0$ m).

1/2 `gt` BSM). Rule 1 (`-shield-mode on`) sets $u_{\text{tb}} = -1.0$ if $\hat{p}_{\text{AEB}} \geq 0.70$, TL = `red`, $d_{\text{ped}} \leq 16.0$ m, or $d_{\text{lead}} \leq 11.5$ m; Rule 2 cancels lane changes if $\max_c \hat{p}_c^{(m)} < 0.80$, `adj_wp.lane_id` $\times$ `wp.lane_id` ≤ 0 [15], or $\text{BS}_{\text{L/R}}$ is latched.

For lane transitions, $N_b = 14$ nodes are sampled 2.0 m apart ahead of the ego axle ($d_0 = 0$ m) at $d_k = 2.0(k+1)$ m for $k \in \{0, \ldots, 13\}$ ($d_0 = 2.0$ m to $d_{13} = 28.0$ m; 26.0 m between P_0 and P_{13}, 28.0 m lookahead from origin, $\tau_k = (k+1)/N_b$) via $s(\tau) = 10\tau^3 - 15\tau^4 + 6\tau^5$ [16, 41]:

$$s(\tau) = 10\tau^3 - 15\tau^4 + 6\tau^5, \tag{11}$$

$$P_k^{\text{blend}} = \big(1 - s(\tau_k)\big) P_k^{\text{orig}} + s(\tau_k) P_k^{\text{adj}}. \tag{12}$$

Envelope $s'(0) = s'(1) = s''(0) = s''(1) = 0$, while pure pursuit projects onto the 2.0 m piecewise-linear polyline $\overline{P_k^{\text{blend}} P_{k+1}^{\text{blend}}}$, bounding discrete heading steps ($|\Delta\psi_k| \leq 1.9°$) rather than analytic C^2 continuity between nodes, with $L_{\text{look}} = \text{clip}(7.5 + 0.18 v_{\text{kmh}}, 7.5, 13.5)$ m [38, 20, 11, 33]. With $\kappa_{\text{map}} = -\Delta\psi_{\text{arc}}/L_{\text{arc}} = -0.045$ `road_curvature` (m^{-1}, $L_{\text{arc}} = 22.2$ m), node $k = 3$ ($t = 1.5$ s, $(\hat{x}_3, \hat{y}_3)$), $L = 2.85$ m, $\delta_{\max} = 0.70$ rad, $k_{\text{FF}} = 0.12$, and CTH-ACC ($d_{\text{lead}} < 38$ m, $v_{\text{lead}} \geq 8$ km/h, $d_0 = 8.0$ m, $\tau_h = 1.2$ s, $k_g = 0.85$ (km/h)/m $\equiv 0.236\,\text{s}^{-1}$)

Table 2: CARLA 0.9.16 (`Town10HD_Opt`) 10 paired episodes (680 m) and 10×5 cross-weather suite (3,400 m) under `vision_estimated` (Vis) vs. `gt` perception.

Scenario	Paired Weather	Event	10-Ep (Vis/GT)	50-Ep (Vis/GT)
FSD-1	ClearNoon	Red-light stop & go	1/1 / 1/1	5/5 / 5/5
FSD-2	CloudyNoon	Quintic left overtake	1/1 / 1/1	5/5 / 5/5
FSD-3	WetNoon	Crosswalk pedestrian AEB	1/1 / 1/1	5/5 / 5/5
FSD-4	SoftRainNoon	CTH-ACC (24 km/h, 16 m)	1/1 / 1/1	5/5 / 5/5
FSD-5	ClearSunset	Quintic right overtake	1/1 / 1/1	5/5 / 5/5
FSD-6	WetCloudyNoon	Lead stop-and-go recovery	1/1 / 1/1	5/5 / 5/5
FSD-7	MidRainyNoon	Front-lat latch (3/3) vs. rear approach (0/2 Vis)	1/1 / 1/1	3/5 / 4/5
FSD-8	WetSunset	Workzone right bypass	1/1 / 1/1	5/5 / 5/5
FSD-9	ClearNight	Night headlight ped AEB	1/1 / 1/1	4/5 / 5/5
FSD-10	HardRainNoon	Heavy-rain cornering ACC	1/1 / 1/1	5/5 / 5/5

[22, 35, 39]:

$$\kappa_{\text{traj}} = \frac{2(\hat{y}_3 - e_y)}{\max(3.5, \hat{x}_3)^2 + (\hat{y}_3 - e_y)^2}, \tag{13}$$

$$\kappa_{\text{eff}} = 0.70(-0.045\,\texttt{road_curvature}) + 0.30\kappa_{\text{traj}}, \tag{14}$$

$$\delta_{\text{raw}} = \text{clip}\left(\delta_{\text{PID}}(e_y, e_\psi) + k_{\text{FF}} \frac{\arctan(L\kappa_{\text{eff}})}{\delta_{\max}}, -1, 1\right), \tag{15}$$

$$\delta_t = \delta_{t-1} + \text{clip}\big(\alpha(\delta_{\text{raw}} - \delta_{t-1}), -\Delta\delta_{\max}, \Delta\delta_{\max}\big), \tag{16}$$

$$d^*(v_{\text{ego}}) = d_0 + \tau_h(v_{\text{ego}}/3.6), \; v_{\text{raw}} = v_{\text{lead}} + k_g(d_{\text{lead}} - d^*), \tag{17}$$

$$v_{\text{ACC}} = \text{clip}\big(v_{\text{raw}}, 0.0, \min(35.0, v_{\text{lead}} + 6.0)\big). \tag{18}$$

```
# scripts/04_evaluate_sim.py & src/control/shield_pid.py
if lead_d_window and abs(pred_d - lead_d_window[-1]) > 6.0:
    lead_d_window = [pred_d]
dd_dt_fused = 0.70 * dd_dt_reg + 0.30 * pred_rr_mps
raw_v_lead = ego_v_kmh + 3.6 * dd_dt_fused
est_v_lead = float(np.clip(0.65*prev_v + 0.35*raw_v_lead, 0, 60))
d_star = 8.0 + 1.2 * (current_speed_kmh / 3.6)
acc_cap = min(35.0, lead_speed_kmh + 6.0)
raw_acc = lead_speed_kmh + 0.85 * (lead_distance_m - d_star)
acc_sync_speed = float(np.clip(raw_acc, 0.0, acc_cap))
traj_curv = (2.0 * ly) / (lx * lx + ly * ly)
eff_curv = 0.70 * (-road_curv * 0.045) + 0.30 * traj_curv
```

Table 3: Closed-loop evaluation separating zero-actor-query `vision_estimated` perception from simulator `gt` state (`checkpoints/eval_metrics_carla.json`).

Simulator	Metric	Target KPI	`vision_estimated` Mode (No Sim Actor Query)	`gt` State Hybrid Mode	Status
CARLA 0.9.16	Single-Frame Latency (32.40 ms neural + 0.45 ms ctrl; +6.40 ms sim IPC)	≤ 40.0 ms	32.85 ms (30.4 FPS, P95: 33.96 ms)	32.85 ms (31.67 ms w/o aux+traj)	PASS
	`Town10HD_Opt`: 10 Paired Eps (680 m) / 50 Cross-Weather (3,400 m)	$\geq 85.0\%$	100.0% (10/10) / 94.0% (47/50, 3-seed 46.7 ± 0.6)	100.0% (10/10) / 98.0% (49/50)	PASS
	`Town10HD_Opt`: 10 Multi-Block Long Routes (200 m/ep, 2,000 m)	$\geq 85.0\%$	90.0% (9/10, $R_c = 95.4\%$)	100.0% (10/10, $R_c = 100.0\%$)	PASS
	`Town03` Unseen Map (10×200 m Roundabout/Grade, 2,000 m)	$\geq 80.0\%$	80.0% (8/10, $R_c = 91.2\%$, $\overline{\lvert e_y \rvert} = 0.112$ m)	90.0% (9/10, $R_c = 96.5\%$)	PASS
	Stress Test: 10% Frame Drop + 25% Patch Occlusion (50-Ep)	$\geq 80.0\%$	88.0% (44/50, $\Delta V_t/\Delta n_t$ + latch active)	94.0% (47/50)	PASS
	Teacher ECE_{mean} / Physical ECE_{phys} / $\text{HCER}_{0.90,\text{AEB}}$ / Physical FNR_{AEB}	≤ 0.0500	0.0142 / 0.0318 / 0.45% (1.12% all) / 1/48 (2.1%)	0.0000 / 0.0110 / 0.10% / 0/48	PASS
	Lead Depth MAE / Co-Pres Ped MAE / Speed Jitter RMS / $\overline{\lvert e_y \rvert}$	≤ 5.0 m / ≤ 0.25 m	3.23 m / 2.64 m / 1.84 km/h / 0.094 m	0.0 m / 0.0 m / 0.0 km/h / 0.085 m	PASS
MetaDrive 0.4.3 ($\Delta t = 0.05$ s)	Single-Frame Latency ($\bar{v} = 33.0$ km/h, 190.4 m & 415.3 steps/ep)	≤ 40.0 ms	30.20 ms (33.1 FPS, P95: 31.67 ms)	30.20 ms (33.1 FPS)	PASS
	Arrival: 10-Ep (4,153 steps, 1,904 m) / 50-Ep (20,765 steps, 9,520 m)	$\geq 85.0\%$	100.0% (10/10) / 96.0% (48/50, CI [86.3, 99.5]%)	100.0% (10/10) / 100.0% (50/50; 98.7% arc)	PASS
	Unseen Closed-Loop Test ECE_{mean} / NLL / Brier ($M = 15$ Bins)	≤ 0.0500	0.0184 / 0.096 / 0.031 (Noisy Sensor)	0.0000 / 0.041 / 0.012 (GT Teacher)	PASS

Table 4: CARLA 0.9.16 ablations: 3-seed mean ± std and paired two-sided McNemar p-values [31] vs. `vision_estimated` DINOv2+Laya (Part A); hybrid trajectory/controller ablations (Part B). Continuous and 10-Ep AEB metrics are from the 10-Ep paired suite (5 hazard events; AEB event = ≥ 3 contiguous ticks with $u_{\text{tb}} \leq -0.50$ bridging ≤ 2-tick gaps; 1–2 tick transients count in ticks only; Full 44 TP ticks = 39 model $\hat{p}_{\text{AEB}} \geq 0.70 + 5$ shield triggers).

Part A: Decision & Backbone (Fixed Controller)	10-Ep	50-Ep Vis (3-Seed / McNemar p)	200 m	$\overline{\lvert e_y \rvert}$	$\lvert \Delta\hat{\tau} \rvert_{\text{RMS}}$	10-Ep TP/FP Evt (Ticks)
Keep-Lane Rule (`rule_only`, `gt`)	6/10	30/50 (60.0%, b=17, c=0, p=10^{-5})	5/10	0.142	0.0195	5/3 (44/14)
Telemetry-Prompt Only ($\hat{V}_t$ = **0**, `vision_est`)	7/10	33/50 (32.7±0.6, b=14, c=0, p=0.0001)	6/10	0.154	0.0298	4/5 (38/26)
DINOv2 + Random Frozen Laya (395.21M raw, `via`)	8/10	36/50 (35.7±1.2, b=11, c=0, p=0.0010)	6/10	0.136	0.0281	4/4 (41/21)
Direct `ExpertLabeler` Rules (`vision_estimated`)	8/10	38/50 (76.0%, b=9, c=0, p=0.0039)	7/10	0.131	0.0296	5/4 (43/18)
DINOv2 + 2-Layer MLP Head (No 395.21M Laya, `via`)	8/10	39/50 (38.7±0.6, b=8, c=0, p=0.0078)	7/10	0.118	0.0265	5/3 (42/14)
Heuristic Overtake FSM (`heuristic_fsm`, `gt`)	8/10	39/50 (78.0%, b=8, c=0, p=0.0078)	7/10	0.126	0.0284	5/4 (44/20)
DINOv2 + 2-Layer Trainable Transformer (7.4M, `via`)	9/10	42/50 (41.7±0.6, b=5, c=0, p=0.0625)	8/10	0.106	0.0249	5/2 (44/9)
DINOv2+Laya w/o Temporal ΔV_t (`no_temporal`, `via`)	9/10	44/50 (43.7±0.6, b=3, c=0, p=0.2500)	8/10	0.102	0.0244	5/2 (44/8)
DINOv2+Laya + 100 ms Intrs-Policy Lag (2-frame)	9/10	43/50 (86.0%, b=4, c=0, p=0.1250)	8/10	0.115	0.0258	5/2 (44/10)
DINOv2+Laya + 300 ms Intrs-Policy Lag (6-frame)	7/10	34/50 (68.0%, b=13, c=0, p=0.0002)	6/10	0.168	0.0315	4/6 (39/33)
DINOv2+Laya (`vision_estimated`, Shield OFF)	9/10	43/50 (42.7±0.6, b=4, c=0, p=0.1250)	8/10	0.098	0.0241	4/1 (42/2)
DINOv2+Laya (`vision_estimated`, Shield ON)	10/10	47/50 (46.7±0.6, Reference)	9/10	0.094	0.0231	5/1 (45/4)
DINOv2+Laya (`gt` State, Shield OFF, `model_only`)	10/10	48/50 (47.7±0.6, b=0, c=1, p=1.0000)	9/10	0.086	0.0226	5/0 (44/1)
DINOv2+Laya (`gt` State, Shield ON, Full)	10/10	49/50 (48.7±0.6, b=0, c=2, p=0.5000)	10/10	0.085	0.0223	5/0 (44/2)
Part B: Hybrid Trajectory & Controller (10-Ep GT / Vis)	**10-Ep GT**	**50-Ep Vis**	TTC_{min} (GT/Vis)	$J_{\text{e,RMS}}$ (GT/Vis)	$\overline{\lvert e_y \rvert}$ (GT/Vis)	10-Ep GT TP/FP Evt (Ticks)
Full Hybrid Stack ($0.7\hat{w}_{\text{map}} + 0.3\hat{w}_{\text{traj}}$)	10/10	47/50 (94.0%)	2.45/2.38 s	1.14/1.21	0.085/0.094	5/0 (44/2)
Kinematic Traj Prior Only ($\Delta\hat{\tau} = \mathbf{0}$, no $\hat{w}_{\text{traj}} \hat{H}$)	10/10	45/50 (90.0%)	2.40/2.31 s	1.26/1.35	0.090/0.101	5/1 (44/4)
HD-Map Curvature Only ($\kappa_{\text{traj}} = 0$, no Traj Head)	10/10	44/50 (88.0%)	2.38/2.28 s	1.32/1.41	0.092/0.105	5/1 (44/4)
Learned Traj Curvature Only ($\kappa_{\text{map}} = 0, \kappa_{\text{off}} = \kappa_{\text{traj}}$)	9/10	41/50 (82.0%)	2.15/2.04 s	1.74/1.88	0.118/0.124	5/2 (44/9)
– True CTH-ACC (Reverted to Fixed 16 m Gap)	10/10	42/50 (84.0%)	1.42/1.34 s	1.48/1.56	0.095/0.108	5/14 (46/66)
– 28 m Spatial Quintic Spline (3.5 m Step WPs)	10/10	41/50 (82.0%)	1.95/1.86 s	3.65/3.82	0.148/0.162	5/10 (45/49)
– Speed-Damped Linear Steer Ramp (Unramped PID)	10/10	39/50 (78.0%)	1.82/1.74 s	4.41/4.60	0.184/0.198	5/15 (45/73)
All Removed (Discrete 2 m Step WPs + Fixed-Gap ACC)	10/10	35/50 (70.0%)	1.18/1.10 s	4.82/5.04	0.218/0.236	5/19 (46/92)

4 Experiments, Latency Breakdown, and Limitations

Runs used an AMD Ryzen 5 7500F (6C/12T) and RTX 4070 Ti (12 GB) in MetaDrive 0.4.3 [27] ($\Delta t = 0.05$ s, `decision_repeat=5`; 415.3 steps / 190.4 m per run at $\bar{v} = 33.0$ km/h, 20,765 steps / 9,520 m over $N = 50$ unseen maps) and CARLA 0.9.16 [14] (`Town10HD_Opt` and `Town03`, 20 Hz A* `GlobalRoutePlanner` [19]). Both share identical 196×392 crop/resize, frozen DINOv2+Laya, `projector_best.pt`, $(0.05, 0.05, 0.10)$ temperatures, CTH-ACC, and 14-node quintic code without retraining; only vertical FOV (90° vs. 65°) and map APIs differ. Stage timings across 1,981 synchronized frames (`checkpoints/eval_metrics_carla.json`):


```
{"simulator": "carla", "ablation_mode": "full", "episodes": 10,
 "total_frames": 1981, "mean_latency_ms": 32.85,
 "mean_neural_latency_ms": 32.40, "base_stack_latency_ms": 31.67,
 "sim_tick_ipc_transport_ms": 6.40, "p95_latency_ms": 33.96,
 "open_loop_traj_collision_rate_pct": 3.8,
 "counterfactual_obstacle_shift_m": {
   "within_lane_keep_lane_dy6_m": 0.12,
   "lane_change_triggered_dy6_m": 3.41},
 "vision_aux_lead_mae_m": 3.23, "vision_ped_copres_mae_m": 2.64,
 "vision_lead_speed_5step_ols": {"window_steps": 5,
   "jitter_rms_kmh": 1.84, "lead_speed_mae_kmh": 2.41,
   "cutin_recovery_s": 0.15, "braking_lag_s": 0.10},
 "physical_hazard_calibration": {"teacher_ece": 0.0142,
   "physical_ece": 0.0318, "hcer_090_aeb_pct": 0.45},
 "stage_latency_breakdown_ms": {
   "preprocess_crop_resize": 0.42, "dinov2_vit_b14_frozen": 16.43,
   "spatial_projector_core_and_temporal": 0.82,
   "auxiliary_perception_head": 0.49,
   "laya_modernbert_and_discrete_heads": 13.55,
   "trajectory_head_6wp": 0.69,
   "safety_shield_bicycle_ff_pid": 0.45}}
```


All seven stages sum to 32.85 ms (32.40 ms neural + 0.45 ms control), and dropping `auxiliary_perception_head` (0.49 ms) + `trajectory_head_6wp` (0.69 ms) gives 31.67 ms base (39.25 ms loop with 6.40 ms CARLA IPC < 50 ms). In Table 4, an AEB event is ≥ 3 contiguous ticks (≥ 0.15 s) with $u_{\text{tb}} \leq -0.50$ (bridging ≤ 2-tick gaps; 1–2 tick transients count in ticks only, so '5/0 (44/2)' in Full has 2 isolated FP ticks and 0 FP events, and 44 TP ticks = 39 model $\hat{p}_{\text{AEB}} \geq 0.70$ ticks + 5 shield triggers). Two-sided McNemar [31] tests give $p < 0.01$ for DINOv2+Laya (47/50) over `mlp_head` ($39/50, p = 0.0078$), `expert_rule_policy` ($38/50, p = 0.0039$), `random_laya` ($36/50, p = 0.0010$), and `prompt_only` ($33/50, p = 0.0001$); 47/50 vs. 42/50 over the 2-layer Transformer ($b = 5, c = 0$, $p = 0.0625$ two-sided, $p = 0.0313$ one-sided) is an observed edge on night/rain episodes, and dropping $\Delta\hat{\tau}$ (45/50), κ_{traj} (44/50), or κ_{map} (41/50) confirms the hybrid design.

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